Subject: Mathematics | **PERIODIC COMPREHENSIVE TEST – 10** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 6: Triangles** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Marks are indicated against each question.
2. **Section A** comprises 6 questions of 1 mark each (5 MCQs and 1 Assertion-Reason).
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer (LA) questions of 5 marks each.
6. **Section E** comprises 1 Case Study based integrated unit of assessment of 4 marks.
7. Use of calculators is strictly prohibited. Draw neat, labeled geometric diagrams wherever required.

SECTION A: OBJECTIVE QUESTIONS (6 Marks)

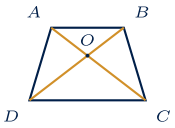
1. If $\Delta ABC \sim \Delta PQR$ with $\angle A = 50°$ and $\angle B = 70°$, then the measure of $\angle R$ is:

(a) $50°$ (b) $70°$ (c) $60°$ (d) $80°$

2. In ΔABC, $AB = 6\sqrt{3}$ cm, $AC = 12$ cm, and $BC = 6$ cm. The angle B is:

(a) $120°$ (b) $60°$ (c) $90°$ (d) $45°$

3. In a trapezium $ABCD$, $AB \parallel CD$ and diagonals intersect at O. If $AB = 6$ cm and $CD = 3$ cm, then the ratio $AO : OC$ is:



(a) $1 : 2$ (b) $2 : 1$ (c) $4 : 1$ (d) $1 : 4$

4. In ΔPQR, S and T are points on PQ and PR such that $\frac{PS}{SQ} = \frac{PT}{TR}$ and $\angle PST = \angle PRQ$. Then ΔPQR is:

(a) Equilateral (b) Isosceles (c) Right-angled (d) Scalene

5. Corresponding sides of two similar triangles are in the ratio $2 : 3$. The ratio of their perimeters is:

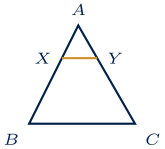
(a) $4 : 9$ (b) $2 : 3$ (c) $3 : 2$ (d) $\sqrt{2} : \sqrt{3}$

6. **Assertion (A):** The shadow of a vertical pole of height 6 m is 4 m. At the same time, the shadow of a tower is 28 m. The height of the tower is 42 m.
Reason (R): At any given instant, the sun's elevation angle is constant, making vertical objects and their shadows form similar triangles.

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(c) Assertion (A) is true but Reason (R) is false.
(d) Assertion (A) is false but Reason (R) is true.

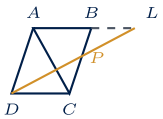
SECTION B: VERY SHORT ANSWER QUESTIONS (4 Marks)

7. **[2 Marks]** In the given figure, $XY \parallel BC$ in ΔABC such that $AX = 2$ cm, $XB = 4$ cm, and $AY = 3$ cm. Find the lengths of YC and AC.



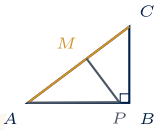
8. **[2 Marks]** Prove that two triangles similar to the same triangle are similar to each other.

SECTION C: SHORT ANSWER QUESTIONS (6 Marks)



9. **[3 Marks]** In the given figure, $ABCD$ is a parallelogram. A line drawn through vertex D intersects side BC at P and ray AB produced at L. Prove mathematically that:

$$\Delta DPC \sim \Delta LPB \quad \text{and} \quad \frac{DP}{PL} = \frac{DC}{BL}$$

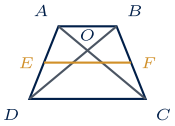


10. **[3 Marks]** In the given figure, ΔABC and ΔAMP are two right triangles, right-angled at B and M respectively. Prove mathematically that:

(i) $\Delta ABC \sim \Delta AMP$

(ii) $\frac{CA}{PA} = \frac{BC}{MP}$

SECTION D: LONG ANSWER / HOTS QUESTIONS (10 Marks)

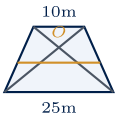


11. **[5 Marks]** In a trapezium $ABCD$, $AB \parallel DC$ and diagonals intersect at O. A line segment EF is drawn parallel to AB passing through O meeting AD at E and BC at F. Prove that: (i) $EO = OF$, (ii) if $AB = a$ and $CD = b$, $EF = \frac{2ab}{a+b}$.

12. **[5 Marks]** In ΔABC, D is a point on BC such that $\angle ADC = \angle BAC$. Prove that $\Delta ABC \sim \Delta DAC$ and $CA^2 = CB \times CD$. If $CB = 16$ cm and $CD = 9$ cm, find CA and the perimeter of ΔDAC given that the perimeter of ΔABC is 48 cm.

SECTION E: COMPETENCY / CASE STUDY QUESTION (4 Marks)

13. **Case Study 10: Civil Engineering Trapezoidal Embankment Construction**



In a highway flyover project, a cross-section of an earth embankment forms a trapezium $ABCD$ with parallel horizontal road layers $AB \parallel DC$. The top width $AB = 10$ m and bottom base $CD = 25$ m. Diagonals AC and BD intersect at drainage culvert O. A reinforcement geogrid layer EF is laid horizontally through O parallel to AB, intersecting side slopes AD at E and BC at F.

(i) **[1 Mark]** Write the ratio in which culvert O divides the diagonal AC.

(ii) **[1 Mark]** Find the ratio $\frac{AE}{AD}$ along the side embankment slope.

(iii) **[2 Marks]** Calculate the total width of the reinforcement geogrid layer EF.

★★★ END OF QUESTION PAPER · ALL THE BEST ★★★